Design and Implementation of a RISC-V Based SoC Subsystem with AXI Interconnect for Embedded Applications

1. INTRODUCTION

1.1 Introduction to Physical Design Flow of RISC-V(PicoRV32) based SoC subsystem architecture

A RISC-V based System-on-Chip featuring the PicoRV32 processor core, AXI-Lite interconnect, ROM, SRAM, and UART peripheral was physically implemented using OpenLane's open-source RTL-to-GDSII flow. The process kicked off with Yosys performing logic synthesis to map the Verilog RTL to SkyWater 130nm standard cells, followed by OpenROAD handling the full backend flow—floorplanning to set die area and power grid, IO placement for clock/reset/UART ports, global and detailed placement of standard cells, clock tree synthesis for balanced skew, and two-stage routing (global then detailed) across all available metal layers.

After routing, OpenSTA verified timing closure with SPEF parasitics at the typical corner, while Magic handled DRC and Netgen performed LVS checks. The final GDSII file emerged ready for fabrication, representing the complete physical layout using Klayout. This compact RISC-V SoC successfully navigated timing, area, and foundry rule constraints through OpenLane's automated yet highly capable flow.

1.2 Objectives

1. To design and implement a RISC-V based SoC subsystem using the PicoRV32 core with AXI-Lite interconnect and peripherals.
2. To perform logic synthesis of the RTL design using Yosys and map it to standard cell libraries.
3. To execute the complete physical design flow (floorplanning, placement, CTS, and routing) using OpenROAD within the OpenLane framework.
4. To validate the design through timing analysis and physical verification using OpenSTA, Magic VLSI, and KLayout.

5. To generate the final GDSII layout and analyze power, performance, and area (PPA) for embedded system applications.

1.3 Block Diagram

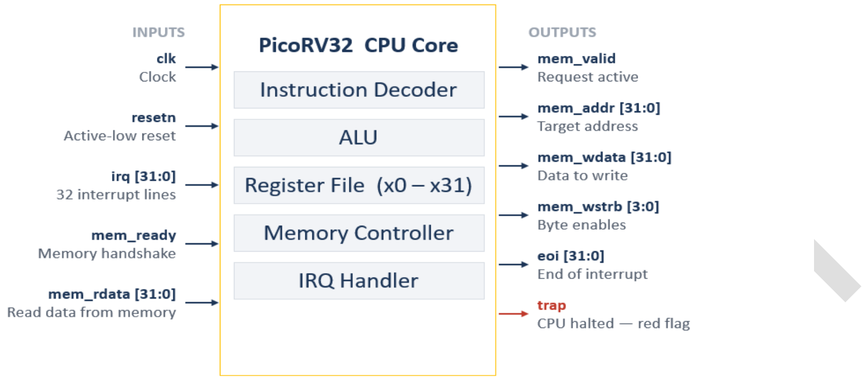

Fig. 1.2.1: Block Diagram of PicoRV32 CPU Core

The PicoRV32 CPU Core consists of modules like the Instruction Decoder, ALU, Register File (x0–x31), Memory Controller, and IRQ Handler, which together execute instructions. Inputs such as clk, resetn, irq [31:0], mem_ready, and mem_rdata [31:0] control operation and provide data. The core generates outputs like mem_valid, mem_addr [31:0], mem_wdata [31:0], and mem_wstrb [3:0] to interact with memory. It also signals eoi [31:0] for interrupt completion and trap to indicate the CPU has halted.

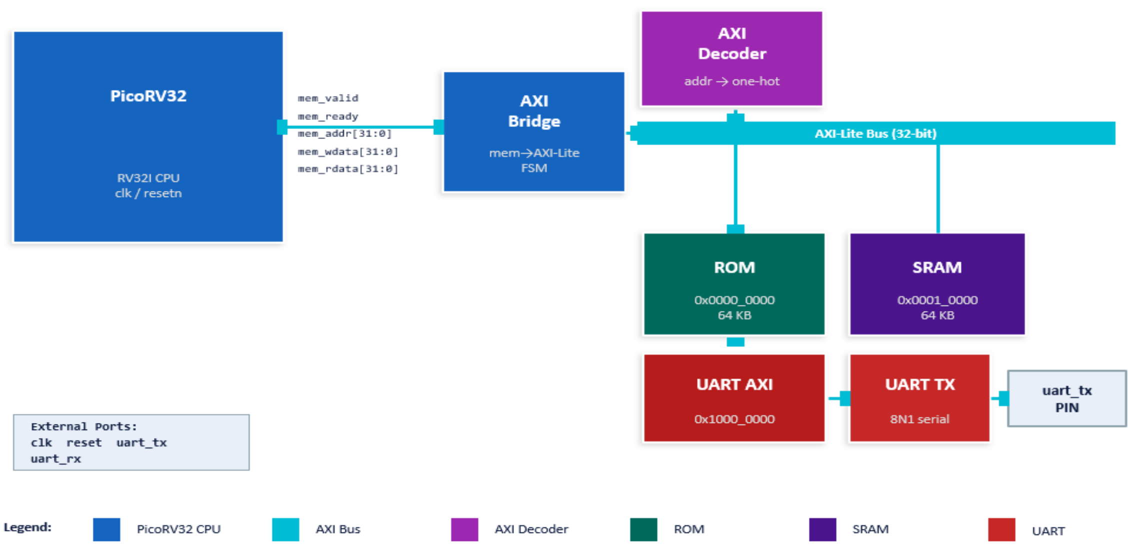

Fig. 1.2.2: Block diagram of the RISC-V(PicoRV32) based SoC subsystem architecture

The PicoRV32 CPU core is synthesized into standard cells and placed with careful consideration of timing paths driven by clk and controlled by resetn. The memory interface signals (mem_valid, mem_ready, mem_addr [31:0], mem_wdata [31:0], mem_rdata [31:0]) are routed to the AXI Bridge, which is implemented as an FSM and optimized for timing and area. The AXI Decoder and AXI-Lite Bus (32-bit) require structured floorplanning to ensure efficient routing and minimal congestion between modules. Memory blocks like ROM and SRAM are typically implemented as macros and placed strategically to reduce wirelength and access latency. The UART AXI and UART TX blocks are placed near I/O pads to simplify routing to the uart_tx pin. Proper clock tree synthesis ensures low skew across all sequential elements, while power distribution and routing are optimized to maintain signal integrity and meet timing closure.

2. SETTING THE ENVIRONMENT IN OPENLANE

Step 1: Set up Project Directory

- Create a directory for your lab: `mkdir <project_name>, mkdir top`

- Open lab directory: `cd <project_name>, cd top`

Step 2: Create Design file

- Create a folder: `mkdir src`
- Open folder: `cd src`

  Inside this folder create `<design>.v` files.

- Open vi/gvim editor for writing RTL design with command: `vi <design_name>.v`
- Open vi/gvim editor to writing testbench for RTL design with command:
  `vi <design_name_tb>.v`
- Open vi/gvim editor for writing the .sdc file with command:
  `vi <design_name>.sdc,`
  `vi top.sdc`

```
create_clock -period 20 -name clk [get_ports clk]
set_clock_uncertainty   0.25 [get_clocks clk]
set_clock_transition    0.25 [get_clocks clk]
set_driving_cell \
    -lib_cell sky130_fd_sc_hd__clkbuf_16 \
    -pin X \
    [get_ports clk]
set_input_delay -min -rise 0.0 [get_ports {reset uart_rx}] -
clock clk
set_input_delay -min -fall 0.0 [get_ports {reset uart_rx}] -
clock clk
set_input_delay -max -rise 4.0 [get_ports {reset uart_rx}] -
clock clk
set_input_delay -max -fall 4.0 [get_ports {reset uart_rx}] -
clock clk
set_input_transition -min -rise 0.15 [get_ports {reset
uart_rx}]
set_input_transition -min -fall 0.15 [get_ports {reset
uart_rx}]
set_input_transition -max -rise 0.50 [get_ports {reset
uart_rx}]
set_input_transition -max -fall 0.50 [get_ports {reset
uart_rx}]
set_output_delay -min -rise 0.0 [get_ports uart_tx] -clock clk
set_output_delay -min -fall 0.0 [get_ports uart_tx] -clock clk
set_output_delay -max -rise 4.0 [get_ports uart_tx] -clock clk
set_output_delay -max -fall 4.0 [get_ports uart_tx] -clock clk
set_load -pin_load 0.017 [get_ports uart_tx]
set_max_transition 1.500 [current_design]
set_false_path -from [get_ports reset]
```

- **To create config.json:** `vi config.json`

```
{
     "DESIGN_NAME": "top",
     "VERILOG_FILES": "dir::src/*.v",
     "VERILOG_DEFINES": "SYNTHESIS",
     "RUN_CTS": true,
     "CLOCK_PERIOD": 20,
     "CLOCK_PORT": "clk",
     "CLOCK_NET": "clk",
     "CLOCK_BUFFER_FANOUT": 16,
     "CTS_CLK_BUFFER_LIST": "sky130_fd_sc_hd__clkbuf_4
      sky130_fd_sc_hd__clkbuf_8
      sky130_fd_sc_hd__clkbuf_16",
     "CTS_ROOT_BUFFER": "sky130_fd_sc_hd__clkbuf_16",
     "CTS_MAX_CAP": 0.35,
     "CTS_TARGET_SKEW": 150,
     "RUN_POST_CTS_RESYNTHESIS": 1,
     "MAX_FANOUT_CONSTRAINT": 8,
     "MAX_TRANSITION_CONSTRAINT": 1.5,
     "MAX_CAPACITANCE_CONSTRAINT": 0.5,
     "OUTPUT_CAP_LOAD": 17.653,
    "BASE_SDC_FILE": "dir::top.sdc",
     "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
      "FP_SIZING": "absolute",
      "DIE_AREA": "0 0 1000 1000",
     "FP_CORE_UTIL": 35,
     "FP_ASPECT_RATIO": 1,
     "PL_TARGET_DENSITY": 0.65,
     "PL_RANDOM_GLB_PLACEMENT": false,
     "GPL_CELL_PADDING": 4,
     "FP_PDN_AUTO_ADJUST": false,
     "FP_PDN_VPITCH": 25,
```

```
"FP_PDN_HPITCH": 25,
"FP_PDN_VOFFSET": 5,
"FP_PDN_HOFFSET": 5,
"GRT_ADJUSTMENT": 0.05,
"GRT_OVERFLOW_ITERS": 100,
"GRT_ALLOW_CONGESTION": 1,
"GLB_RT_MAXLAYER": 4,
"RT_MAX_LAYER": "met4",
"DRT_OPT_ITERS": 32,
"DRT_MIN_ACCESS_POINTS": 1,
"DIODE_INSERTION_STRATEGY": 3,
"PL_RESIZER_MAX_WIRE_LENGTH": 600,
"PL_RESIZER_MAX_SLEW_MARGIN": 10,
"PL_RESIZER_MAX_CAP_MARGIN": 10,
"PL_RESIZER_HOLD_SLACK_MARGIN": 0.35,
"PL_RESIZER_SETUP_SLACK_MARGIN": 0.1,
"GLB_RESIZER_MAX_WIRE_LENGTH": 600,
"GLB_RESIZER_HOLD_SLACK_MARGIN": 0.35,
"GLB_RESIZER_SETUP_SLACK_MARGIN": 0.1,
"ROUTING_CORES": 4,
"GRT_REPAIR_ANTENNAS": 1,
"GRT_ANTENNA_ITERS": 10,
"DIODE_ON_PORTS": "both",
"RUN_HEURISTIC_DIODE_INSERTION": 1,
"SYNTH_STRATEGY": "AREA 3",
"SYNTH_SPLITNETS": 1,
"SYNTH_BUFFERING": 1,
"SYNTH_SIZING": 1,
"SYNTH_SHARE_RESOURCES": 1,
"SYNTH_ADDER_TYPE": "YOSYS",
"RUN_FILL_INSERTION": 1,
"RUN_DRC": 1,
"RUN_LVS": 1,
```

```
    "RUN_ANTENNA_CHECK": 1,
    "RUN_SPICE_EXTRACTION": 1,
    "IO_PCT": 0.2,
    "VDD_NETS": "VPWR",
    "GND_NETS": "VGND",
    "SYNTH_USE_PG_PINS_DEFINES": "USE_POWER_PINS",
    "FP_PDN_CORE_RING": 0,
   "RUN_IRDROP_REPORT": 0
}
```

- **To create pin order configuration**: `vi pin_order.cfg`

```
  #N
clk

 #S
reset

 #E

uart_tx

 #W
uart_rx
```

- **Command to invoke the tool:** `openlane`

```
coe-2@ip-10-0-22-84:~/OpenLaneUser/designs/top$ openlane
========================================
 OpenLane started for user: coe-2
 Workspace : /home/coe-2/OpenLaneUser
 PDK Root  : /labroot/openlane_pdks
========================================
OpenLane Container (ff5509f):/openlane$
```

- **To run the flow in interactive mode:** `flow.tcl -interactive`

```
OpenLane Container (ff5509f):/openlane$ flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.
```

- **To prep your design:** `prep -design top`

```
% prep -design top
[INFO]: Using configuration in 'designs/top/config.json'...
[INFO]: PDK Root: /openlane/pdks
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/top/runs/RUN_2026.04.08_04.49.50
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
```

3. LOGIC SYNTHESIS USING YOSYS IN OPENLANE

- **Command to run Yosys:** `run_synthesis`

```
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/synthesis/2-sta.log)...
```

- After running synthesis open the generated run folder to see the reports that are generated.

  `vi 1-synthesis_pre.stat`

```
55. Printing statistics.

=== top ===

   Number of wires:              11508
   Number of wire bits:          15893
   Number of public wires:         250
   Number of public wire bits:    4275
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:              15610
     $_ANDNOT_                    3628
     $_AND_                        339
     $_DFFE_PP0P_                  350
     $_DFFE_PP1P_                    5
     $_DFFE_PP_                   3286
     $_DFF_PP0_                      9
     $_DFF_PP1_                      5
     $_DFF_P_                       59
     $_MUX_                       2993
     $_NAND_                       279
     $_NOR_                        186
     $_NOT_                        199
     $_ORNOT_                      219
     $_OR_                        3360
     $_SDFFCE_PN0P_                 34
     $_SDFFCE_PP0P_                  1
     $_SDFFE_PP0P_                 127
     $_SDFFE_PP1P_                   5
     $_SDFF_PP0_                    60
     $_SDFF_PP1_                     7
     $_XNOR_                       122
     $_XOR_                        337
```

vi 1-synthesis.AREA_0.stat.rpt

```
69. Printing statistics.

=== top ===

   Number of wires:              19196
   Number of wire bits:          19196
   Number of public wires:        3975
   Number of public wire bits:    3975
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:              19193
     sky130_fd_sc_hd__a2111o_2      65
     sky130_fd_sc_hd__a2111oi_2    129
     sky130_fd_sc_hd__a211o_2       88
     sky130_fd_sc_hd__a211oi_2      58
     sky130_fd_sc_hd__a21bo_2       93
     sky130_fd_sc_hd__a21boi_2      17
     sky130_fd_sc_hd__a21o_2       219
     sky130_fd_sc_hd__a21oi_2      224
     sky130_fd_sc_hd__a21oi_4        1
     sky130_fd_sc_hd__a221o_2      189
     sky130_fd_sc_hd__a221oi_2       1
     sky130_fd_sc_hd__a22o_2       364
     sky130_fd_sc_hd__a22oi_2        3
     sky130_fd_sc_hd__a2bb2o_2      12
     sky130_fd_sc_hd__a2bb2o_4       1
     sky130_fd_sc_hd__a311o_2       10
     sky130_fd_sc_hd__a31o_2       162
     sky130_fd_sc_hd__a31oi_2        1
     sky130_fd_sc_hd__a32o_2        87
     sky130_fd_sc_hd__a32oi_2        2
     sky130_fd_sc_hd__a41o_2         3
     sky130_fd_sc_hd__a41oi_2        1
     sky130_fd_sc_hd__nor3b_2        1
     sky130_fd_sc_hd__o2111a_2       2
     sky130_fd_sc_hd__o2111ai_2      2
     sky130_fd_sc_hd__o211a_2       24
     sky130_fd_sc_hd__o211ai_2       7
     sky130_fd_sc_hd__o21a_2        75
     sky130_fd_sc_hd__o21ai_2      200
     sky130_fd_sc_hd__o21ba_2        2
     sky130_fd_sc_hd__o21bai_2       1
     sky130_fd_sc_hd__o221a_2       20
     sky130_fd_sc_hd__o221ai_2      12
     sky130_fd_sc_hd__o22a_2        42
     sky130_fd_sc_hd__o22ai_2        2
     sky130_fd_sc_hd__o2bb2a_2      12
     sky130_fd_sc_hd__o2bb2ai_2      1
     sky130_fd_sc_hd__o311a_2        2
     sky130_fd_sc_hd__o31a_2         6
     sky130_fd_sc_hd__o31ai_2        4
     sky130_fd_sc_hd__o32a_2        25
     sky130_fd_sc_hd__o41a_2         1
     sky130_fd_sc_hd__or2_2        178
     sky130_fd_sc_hd__or2_4         19
     sky130_fd_sc_hd__or2b_2        86
     sky130_fd_sc_hd__or3_2        100
     sky130_fd_sc_hd__or3_4          4
     sky130_fd_sc_hd__or3b_2        19
     sky130_fd_sc_hd__or3b_4         3
     sky130_fd_sc_hd__or4_2        147
     sky130_fd_sc_hd__or4_4         58
     sky130_fd_sc_hd__or4b_2         1
     sky130_fd_sc_hd__xnor2_2       22
     sky130_fd_sc_hd__xor2_2        29

   Chip area for module '\top': 213174.451200
```

4. FLOORPLAN USING OPENROAD IN OPENLANE

- **To run floorplan:** `run_floorplan`

```
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 3988.66 and height 3976.64.
[STEP 4]
[INFO]: Running IO Placement (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/floorplan/4-place_io.log)...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/floorplan/6-pdn.log)...
```

- View Floorplan using klayout

Locate the DEF file

```
designs/top/runs/RUN_2026.03.24_14.12.30/results/floorplan/top
.def
```

Open in klayout in the same directory where top.def is located

```
klayout top.def
```

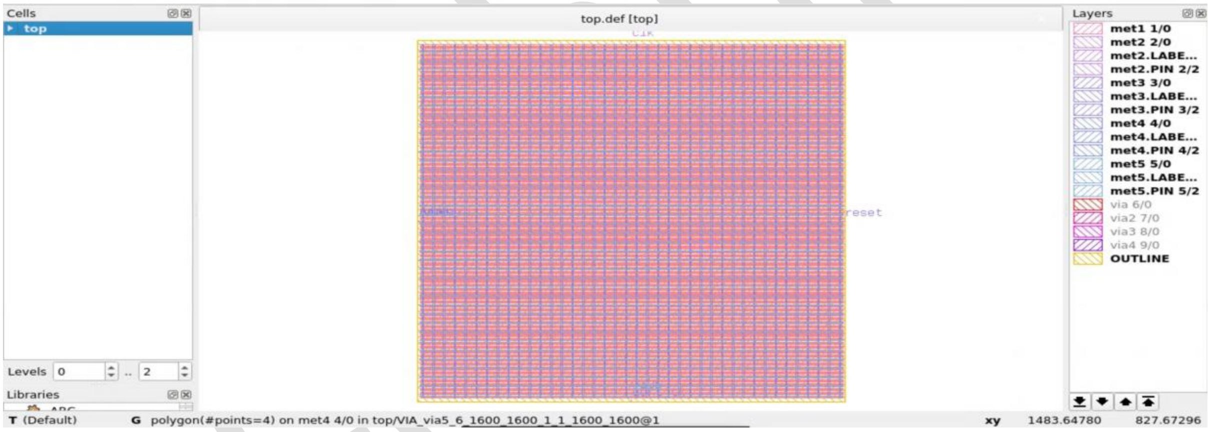

Fig. 4.1: Klayout view of floorplan

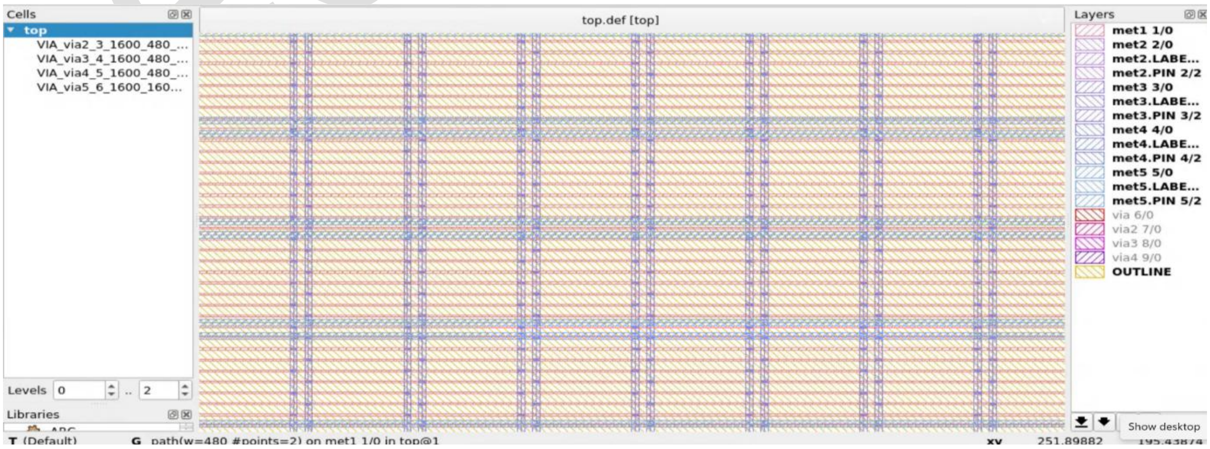

Fig. 4.2: Zoomed klayout view of floorplan

- **To check reports:**

designs/top/runs/RUN_2026.03.24_14.12.30/reports/floorplan/vi initial_fp_core_area.rpt

```
vi 3-initial_fp_core_area.rpt
```

```
5.52 10.88 994.06 987.36
```

designs/top/runs/RUN_2026.03.24_14.12.30/reports/floorplan/vi 3-initial_fp_die_area.rpt

```
vi 3-initial_fp_die_area.rpt
```

```
0.0 0.0 1000.0 1000.0
~
```

5. PLACEMENT USING OPENROAD IN OPENLANE

- **To run placement:** `run_placement`

```
% run_placement
[STEP 7]
[INFO]: Running Global Placement (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/placement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/placement/11-dpl_sta.log)...
```

View placement using klayout:

Locate the DEF file and LEF file, keep them on same folder

- designs/top/runs/RUN_2026.03.24_14.12.30/results/floorplan/top.def
- designs/top/runs/RUN_2026.03.24_14.12.30/tmp/merged.nom.lef

Note: keep both the files (top.def and merged.nom.lef) in the same directory.

```
klayout top.def merged.nom.lef
```

Fig. 5.1: Klayout view of Placement

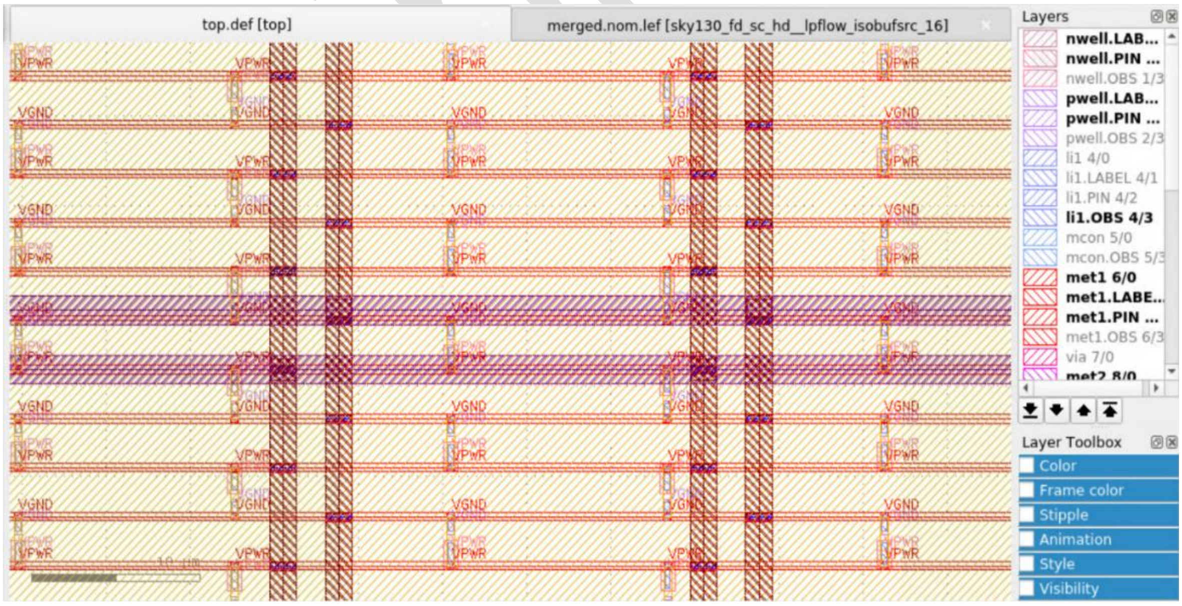

Fig. 5.2: Zoomed Klayout view of Placement

Check for Reports:

To check for STA summary: `vi 10-dpl_sta.summary.rpt`

```
===========================================================================
report_tns
============================================================================
tns 0.00

===========================================================================
report_wns
============================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
============================================================================
worst slack 14.93

===========================================================================
report_worst_slack -min (Hold)
============================================================================
worst slack -0.02
~
~
~
```

6. CLOCK TREE SYNTHESIS USING OPENROAD IN OPENLANE

- **To run CTS**: `run_cts`

```
% run_cts
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/cts/13-cts_sta.log)...
```

Locate .def file and merge.nom.lef file, keep them on same folder and open it using klayout tool

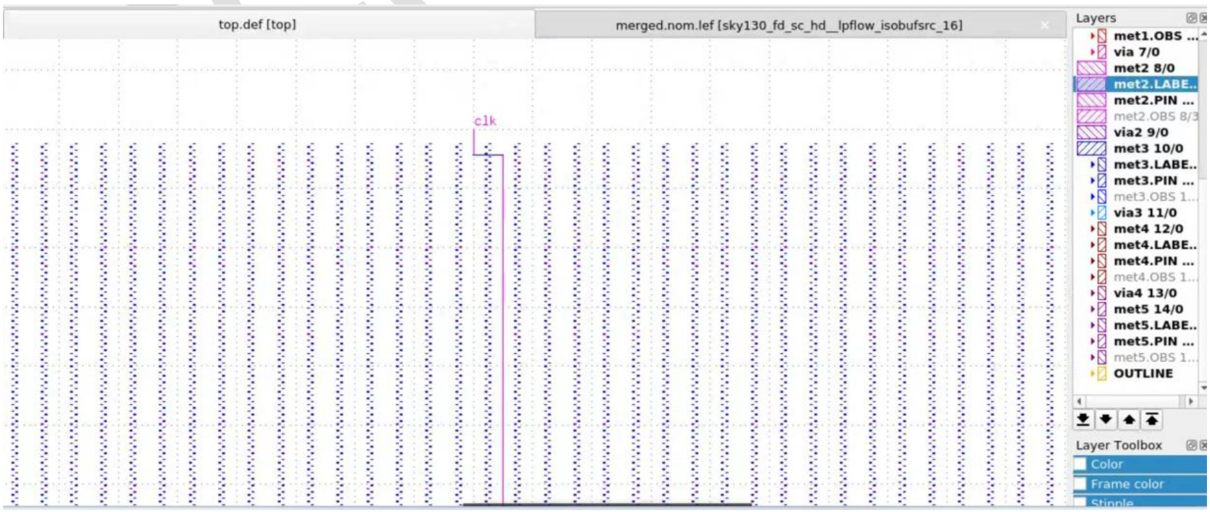

Fig. 6.1: Physical layout view showing the routed clock net (clk)

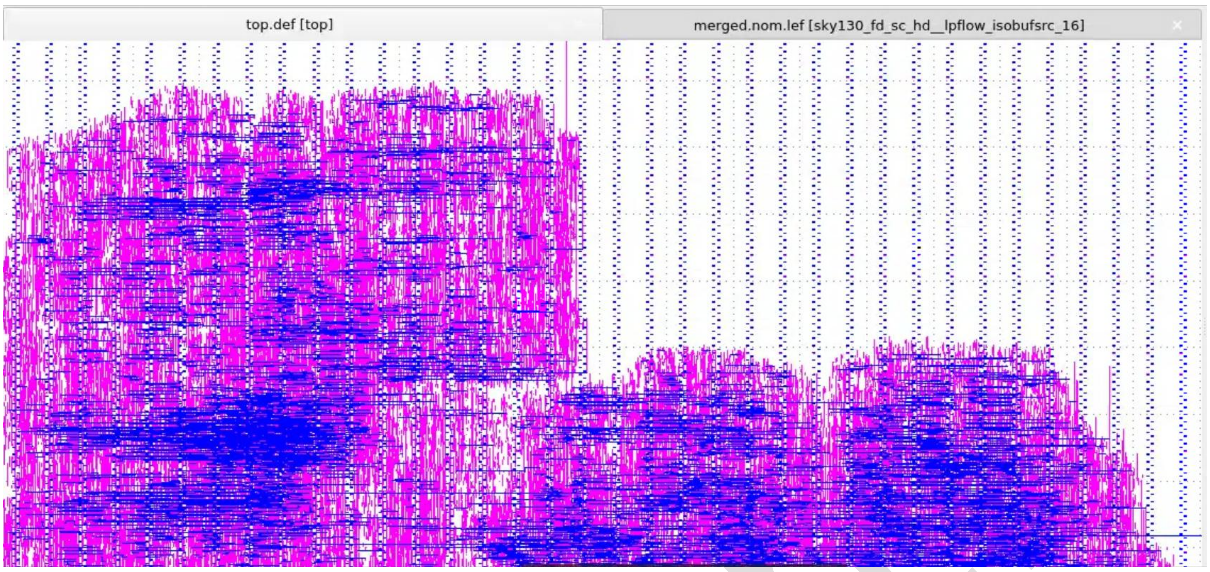

Fig. 6.2: Layout view after the CTS

Check for Reports:-

`/designs/top/runs/RUN_2026.03.24_14.12.30/reports/cts/13-cts_sta.summary.rpt`

To check for STA summary: `vi 13-cts_sta.summary.rpt`

```
===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 14.05

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack -0.90
~
~
```

7. ROUTING USING OPENROAD IN OPENLANE

- **To run routing:** `run_routing`

```
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/14-resizer_design.lo
g)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/16-resizer_timing.lo
g)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running I/O Diode Insertion (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/18-io_diodes.log)...
[STEP 19]
[INFO]: Running Detailed Placement (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/19-io_diode_legalization.log)...
[STEP 20]
[INFO]: Running Heuristic Diode Insertion (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/20-diodes.log)...
[STEP 21]
[INFO]: Running Detailed Placement (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/21-diode_legalization.log)...
[STEP 22]
[INFO]: Running Global Routing (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/22-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 23]
[INFO]: Writing Verilog (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/22-global_write_netlist.log)...
[STEP 24]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/24-grt_sta.log)...
[STEP 25]
[INFO]: Running Fill Insertion (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/25-fill.log)...
[STEP 26]
[INFO]: Running Detailed Routing (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/routing/26-detailed.log)...
[INFO]: No DRC violations after detailed routing.
```

Locate .def file and merge.nom.lef file, keep them on same folder and open it using klayout tool

Fig. 7: klayout view of Routing

Check for reports:

`OpenLaneUser/designs/top/runs/ RUN_2026.03.24_14.12.30 /reports/signoff/35-rcx_sta.summary.rpt`

To check for STA summary: `vi 35-rcx_timing_sta.summary.rpt`

```
===========================================================================
report_tns
============================================================================
tns 0.00

===========================================================================
report_wns
============================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
============================================================================
worst slack 13.78

===========================================================================
report_worst_slack -min (Hold)
============================================================================
worst slack 0.71
~
~
```

8. PARASITIC EXTRACTION IN OPENLANE

To run magic: `run_magic`

```
% run_magic
[STEP 28]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/28-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/28-lef.log)...
```

- **For spice export:** `run_magic_spice_export`

```
% run_magic_spice_export
[STEP 30]
[INFO]: Running Magic Spice Export from LEF (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/30-spice.log)...
```

- **To see spice file:** `vi top.lef.spice`

`designs/top/runs/RUN_2026.03.24_14.12.30/results/signoff/top.lef.spice`

```
NGSPICE file created from top.ext - technology: sky130A

* Black-box entry subcircuit for sky130_ef_sc_hd__decap_12 abstract view
.subckt sky130_ef_sc_hd__decap_12 VPWR VGND VPB VNB
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__diode_2 abstract view
.subckt sky130_fd_sc_hd__diode_2 DIODE VGND VNB VPB VPWR
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__decap_6 abstract view
.subckt sky130_fd_sc_hd__decap_6 VGND VNB VPB VPWR
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__fill_1 abstract view
.subckt sky130_fd_sc_hd__fill_1 VGND VNB VPB VPWR
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__dfxtp_1 abstract view
.subckt sky130_fd_sc_hd__dfxtp_1 CLK D VGND VNB VPB VPWR Q
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__fill_2 abstract view
.subckt sky130_fd_sc_hd__fill_2 VGND VNB VPB VPWR
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__a22o_1 abstract view
.subckt sky130_fd_sc_hd__a22o_1 A1 A2 B1 B2 VGND VNB VPB VPWR X
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__decap_3 abstract view
.subckt sky130_fd_sc_hd__decap_3 VGND VNB VPB VPWR
.ends

* Black-box entry subcircuit for sky130_fd_sc_hd__decap_4 abstract view
"top.lef.spice" [readonly] 136997L, 9114797C
```

9. DESIGN RULE CHECKING USING MAGIC IN OPENLANE

- **For Design Rule Check:** `run_magic_drc`

```
% run_magic_drc
[STEP 31]
[INFO]: Running Magic DRC (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/31-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
```

- **Check for DRC report:**

`vi drc.rpt`

```
OpenLane Container (ff5509f):/openlane/designs/top/runs/RUN_2026.04.08_04.49.50/reports/signoff$ ls
drc.klayout.xml  drc.rdb  drc.rpt  drc.tcl  drc.tr  spice.feedback.txt
```

```
top
----------------------------------------
[INFO]: COUNT: 0
```

10. STREAM OUT GDSII USING KLAYOUT IN OPENLANE

- **To run for klayout:** `run_klayout`

```
% run_klayout
[STEP 32]
[INFO]: Streaming out GDSII with KLayout (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/32-gdsii-klayout.log)...
```

- To view the final

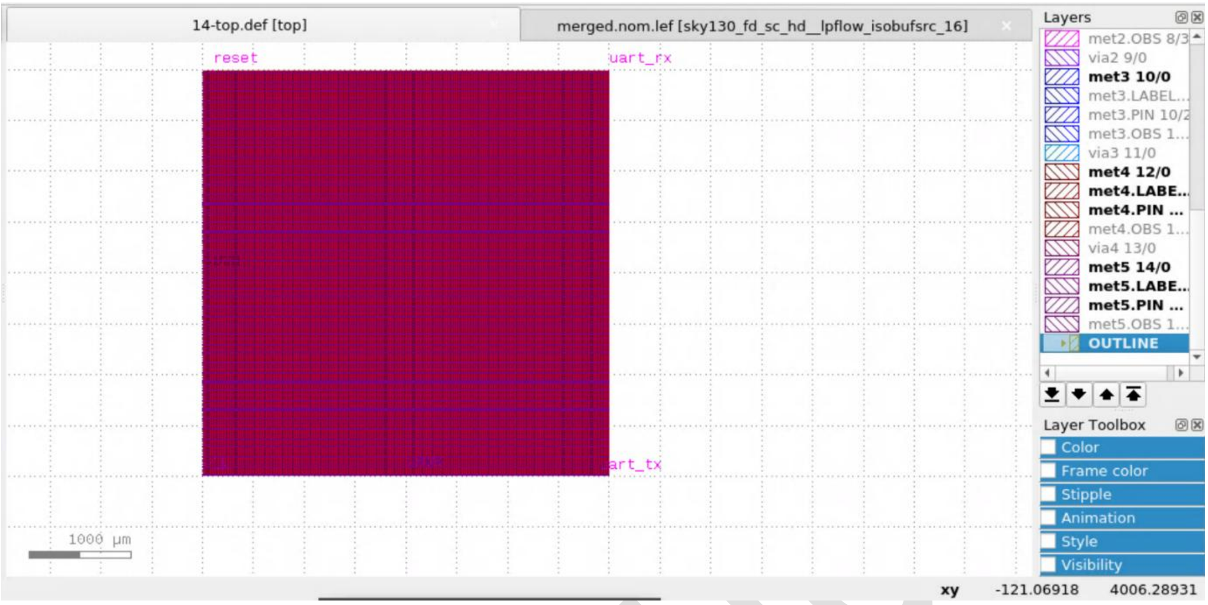

Fig. 10.1: klayout view of final layout

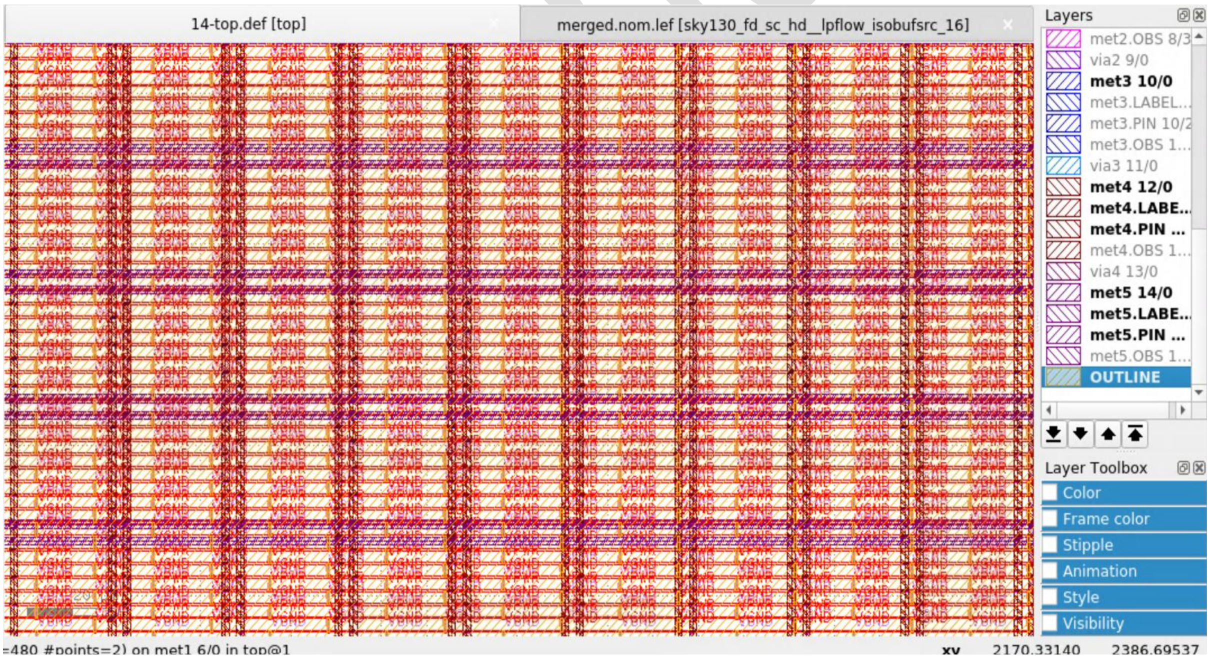

Fig. 10.2: Zoomed view of layout on klayout

11. LAYOUT VERSUS SCHEMATIC CHECK USING KLAYOUT IN OPENLANE

To check for LVS: `run_lvs`

```
% run_lvs
[STEP 33]
[INFO]: Writing Powered Verilog (logs: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/33-write_powered_def.log, designs/top/runs/RUN_
2026.04.08_04.49.50/logs/signoff/33-write_powered_verilog.log)...
[STEP 34]
[INFO]: Writing Verilog (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/33-write_powered_verilog.log)...
[STEP 35]
[INFO]: Running LVS (log: designs/top/runs/RUN_2026.04.08_04.49.50/logs/signoff/35-lvs.lef.log)...
```

```
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
```

Check for LVS reports:

Locate the lvs report

`/OpenLaneUser/designs/top/runs/RUN_2026.03.24_14.12.30/reports/signoff/30-top.lvs.rpt`

Open the report using vim editor

`vi 30-top.lvs.rpt`

```
LVS reports no net, device, pin, or property mismatches.

Total errors = 0
~
~
~
```

Final GDSII file:
`/OpenLaneUser/designs/top/runs/RUN_2026.03.24_14.012.30/results/signoff/top.gds`

- **To run for klayout**: `run_klayout`

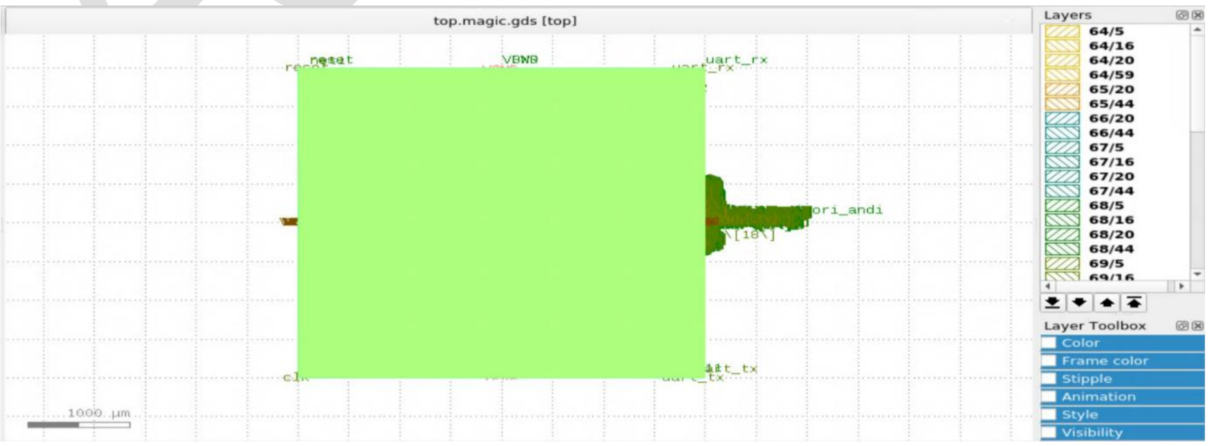

Fig. 11.1: Klayout view of final GDSII

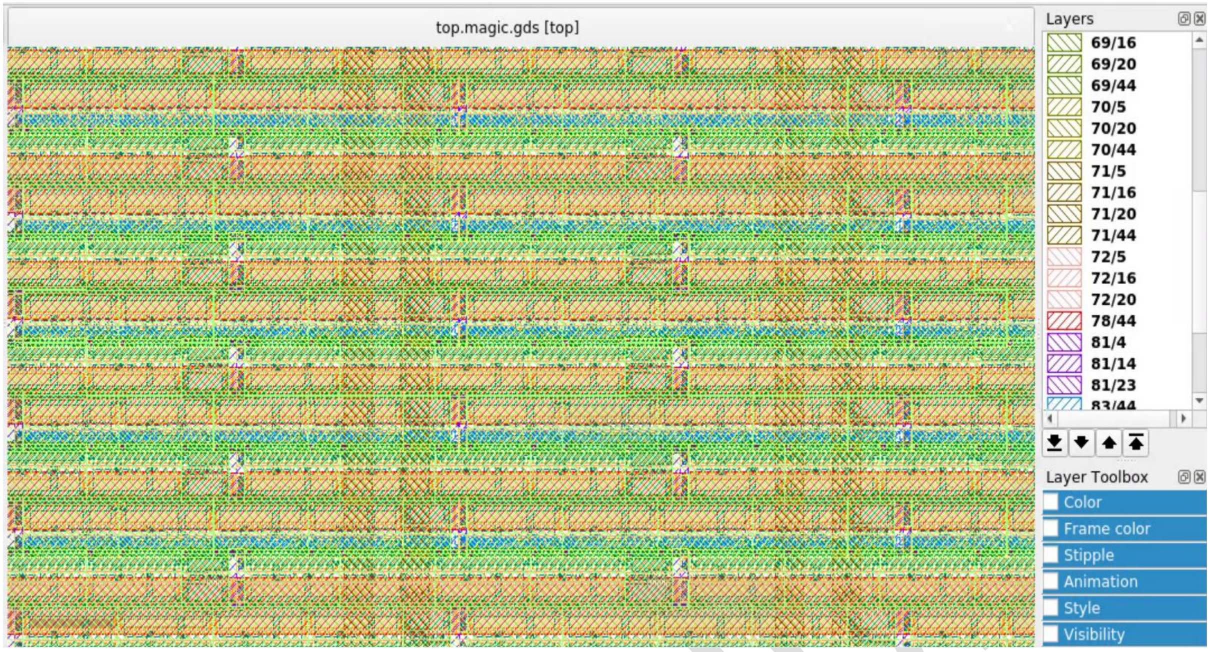

Fig. 11.2: Klayout view of final GDSII

Fig. 11.3: Final GDSII File

12. ANALYSING POWER, PERFORMANCE AND AREA USING DIFFERENT REPORT GENERATED BY OPENLANE.

- Power report after synthesis

```
===========================================================================
 report_power
============================================================================
======================= Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             9.35e-03   6.92e-04   3.42e-08   1.00e-02  51.3%
Combinational          5.83e-03   3.70e-03   4.98e-08   9.53e-03  48.7%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  1.52e-02   4.39e-03   8.39e-08   1.96e-02 100.0%
                          77.6%      22.4%       0.0%
```

- Power report after Routing

```
===========================================================================
 report_power
============================================================================
====================== Typical Corner ==================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             9.17e-03   4.78e-04   3.45e-08   9.65e-03  36.3%
Combinational          1.09e-02   6.05e-03   8.78e-08   1.69e-02  63.7%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  2.00e-02   6.53e-03   1.22e-07   2.66e-02 100.0%
                          75.4%      24.6%       0.0%
```

- Timing report After Synthesis

Setup report

```
Startpoint: _34277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: uart_tx (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout      Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.25    0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (ideal)
                   0.25    0.00    0.00 ^ _34277_/CLK (sky130_fd_sc_hd__dfstp_2)
                   0.10    0.63    0.63 ^ _34277_/Q (sky130_fd_sc_hd__dfstp_2)
     1     0.02                           uart_tx (net)
                   0.10    0.00    0.63 ^ uart_tx (out)
                                   0.63   data arrival time

                   0.25   20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock network delay (ideal)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                          -4.00   15.75   output external delay
                                  15.75   data required time
-----------------------------------------------------------------------------
                                  15.75   data required time
                                  -0.63   data arrival time
-----------------------------------------------------------------------------
                                  15.12   slack (MET)
```

Hold report

```
===========================================================================
report_checks -path_delay min (Hold)
============================================================================
======================= Typical Corner ===================================

Startpoint: uart_rx (input port clocked by clk)
Endpoint: _34190_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout      Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.25    0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (ideal)
                           0.00    0.00 v input external delay
                   0.15    0.00    0.00 v uart_rx (in)
     1     0.00                           uart_rx (net)
                   0.15    0.00    0.00 v _34190_/D (sky130_fd_sc_hd__dfstp_2)
                                   0.00   data arrival time

                   0.25    0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (ideal)
                           0.25    0.25   clock uncertainty
                           0.00    0.25   clock reconvergence pessimism
                                   0.25 ^ _34190_/CLK (sky130_fd_sc_hd__dfstp_2)
                           0.00    0.25   library hold time
                                   0.25   data required time
-----------------------------------------------------------------------------
                                   0.25   data required time
                                  -0.00   data arrival time
-----------------------------------------------------------------------------
                                  -0.25   slack (VIOLATED)
```

- Timing report After CTS

Setup report

```
===========================================================================
report_checks -path_delay max (Setup)
============================================================================
======================= Typical Corner ===================================

Startpoint: _34277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: uart_tx (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.07    0.05    0.05 ^ clk (in)
     1    0.05                           clk (net)
                  0.07    0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.09    0.18    0.23 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.08                           clknet_0_clk (net)
                  0.09    0.00    0.23 ^ clkbuf_2_0_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.09    0.18    0.41 ^ clkbuf_2_0_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.07                           clknet_2_0_0_clk (net)
                  0.09    0.00    0.41 ^ clkbuf_4_2_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.15    0.56 ^ clkbuf_4_2_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.03                           clknet_4_2_0_clk (net)
                  0.05    0.00    0.56 ^ clkbuf_5_5__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.18    0.23    0.79 ^ clkbuf_5_5__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    15    0.16                           clknet_5_5__leaf_clk (net)
                  0.18    0.00    0.79 ^ clkbuf_leaf_369_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.16    0.95 ^ clkbuf_leaf_369_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_leaf_369_clk (net)
                  0.03    0.00    0.95 ^ _34277_/CLK (sky130_fd_sc_hd__dfstp_1)
                  0.14    0.57    1.51 ^ _34277_/Q (sky130_fd_sc_hd__dfstp_1)
     1    0.01                           net3 (net)
                  0.14    0.00    1.51 ^ output3/A (sky130_fd_sc_hd__clkbuf_2)
                  0.10    0.18    1.70 ^ output3/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.02                           uart_tx (net)
                  0.10    0.00    1.70 ^ uart_tx (out)
                                  1.70   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-----------------------------------------------------------------------------
                                 15.75   data required time
                                 -1.70   data arrival time
-----------------------------------------------------------------------------
                                 14.05   slack (MET)
```

Hold report

```
eport_checks -path_delay min (Hold)
==========================================================================
====================== Typical Corner ===================================

tartpoint: _34405_ (rising edge-triggered flip-flop clocked by clk)
ndpoint: _31207_ (rising edge-triggered flip-flop clocked by clk)
ath Group: clk
ath Type: min

anout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                 0.03    0.01    0.01 ^ clk (in)
    3    0.01                           clk (net)
                 0.03    0.00    0.01 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.13    0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    8    0.04                           clknet_0_clk (net)
                 0.06    0.00    0.15 ^ clkbuf_2_2_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.14    0.29 ^ clkbuf_2_2_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    8    0.04                           clknet_2_2_0_clk (net)
                 0.06    0.00    0.29 ^ clkbuf_4_8_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.13    0.42 ^ clkbuf_4_8_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    4    0.02                           clknet_4_8_0_clk (net)
                 0.04    0.00    0.42 ^ clkbuf_5_16__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.13    0.20    0.61 ^ clkbuf_5_16__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
   26    0.11                           clknet_5_16__leaf_clk (net)
                 0.13    0.00    0.61 ^ clkbuf_leaf_312_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.15    0.76 ^ clkbuf_leaf_312_clk/X (sky130_fd_sc_hd__clkbuf_16)
    6    0.01                           clknet_leaf_312_clk (net)
                 0.04    0.00    0.76 ^ _34405_/CLK (sky130_fd_sc_hd__dfxtp_2)
                 0.09    0.35    1.11 ^ _34405_/Q (sky130_fd_sc_hd__dfxtp_2)
   10    0.02                           u_cpu.decoded_imm[7] (net)
                 0.09    0.00    1.11 ^ hold5184/A (sky130_fd_sc_hd__buf_2)
                 0.04    0.12    1.23 ^ hold5184/X (sky130_fd_sc_hd__buf_2)
    4    0.01                           net5190 (net)
                 0.04    0.00    1.23 ^ _21948_/A1 (sky130_fd_sc_hd__a22o_1)
                 0.05    0.12    1.35 ^ _21948_/X (sky130_fd_sc_hd__a22o_1)
    2    0.00                           _08207_ (net)
                 0.05    0.00    1.35 ^ _21949_/A1 (sky130_fd_sc_hd__mux2_1)
                 0.03    0.11    1.46 ^ _21949_/X (sky130_fd_sc_hd__mux2_1)

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                 0.03    0.01    0.01 ^ clk (in)
    3    0.01                           clk (net)
                 0.03    0.00    0.01 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.13    0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    8    0.04                           clknet_0_clk (net)
                 0.06    0.00    0.15 ^ clkbuf_2_1_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.14    0.29 ^ clkbuf_2_1_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    8    0.04                           clknet_2_1_0_clk (net)
                 0.06    0.00    0.29 ^ clkbuf_4_6_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.13    0.42 ^ clkbuf_4_6_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    4    0.02                           clknet_4_6_0_clk (net)
                 0.04    0.00    0.42 ^ clkbuf_5_13__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.17    0.23    0.64 ^ clkbuf_5_13__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
   36    0.16                           clknet_5_13__leaf_clk (net)
                 0.17    0.00    0.64 ^ clkbuf_leaf_172_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.16    0.80 ^ clkbuf_leaf_172_clk/X (sky130_fd_sc_hd__clkbuf_16)
    7    0.01                           clknet_leaf_172_clk (net)
                 0.04    0.00    0.80 ^ _31207_/CLK (sky130_fd_sc_hd__dfxtp_2)
                         0.25    1.05   clock uncertainty
                         0.00    1.05   clock reconvergence pessimism
                        -0.03    1.03   library hold time
                                 1.03   data required time
-----------------------------------------------------------------------------
                                 1.03   data required time
                                -1.46   data arrival time
-----------------------------------------------------------------------------
                                 0.43   slack (MET)
```

Area Report

```
69. Printing statistics.

=== top ===

   Number of wires:              19196
   Number of wire bits:          19196
   Number of public wires:        3975
   Number of public wire bits:    3975
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:              19193
     sky130_fd_sc_hd__a2111o_2      65
     sky130_fd_sc_hd__a2111oi_2    129
     sky130_fd_sc_hd__a211o_2       88
     sky130_fd_sc_hd__a211oi_2      58
     sky130_fd_sc_hd__a21bo_2       93
     sky130_fd_sc_hd__a21boi_2      17
     sky130_fd_sc_hd__a21o_2       219
     sky130_fd_sc_hd__a21oi_2      224
     sky130_fd_sc_hd__a21oi_4        1
     sky130_fd_sc_hd__a221o_2      189
     sky130_fd_sc_hd__a221oi_2       1
     sky130_fd_sc_hd__a22o_2       364
     sky130_fd_sc_hd__a22oi_2        3
     sky130_fd_sc_hd__a2bb2o_2      12
     sky130_fd_sc_hd__a2bb2o_4       1
     sky130_fd_sc_hd__a311o_2       10
     sky130_fd_sc_hd__a31o_2       162
     sky130_fd_sc_hd__a31oi_2        1
     sky130_fd_sc_hd__a32o_2        87
     sky130_fd_sc_hd__a32oi_2        2
     sky130_fd_sc_hd__a41o_2         3
     sky130_fd_sc_hd__a41oi_2        1
     sky130_fd_sc_hd__nor3b_2        1
     sky130_fd_sc_hd__o2111a_2       2
     sky130_fd_sc_hd__o2111ai_2      2
     sky130_fd_sc_hd__o211a_2       24
     sky130_fd_sc_hd__o211ai_2       7
     sky130_fd_sc_hd__o21a_2        75
     sky130_fd_sc_hd__o21ai_2      200
     sky130_fd_sc_hd__o21ba_2        2
     sky130_fd_sc_hd__o21bai_2       1
     sky130_fd_sc_hd__o221a_2       20
     sky130_fd_sc_hd__o221ai_2      12
     sky130_fd_sc_hd__o22a_2        42
     sky130_fd_sc_hd__o22ai_2        2
     sky130_fd_sc_hd__o2bb2a_2      12
     sky130_fd_sc_hd__o2bb2ai_2      1
     sky130_fd_sc_hd__o311a_2        2
     sky130_fd_sc_hd__o31a_2         6
     sky130_fd_sc_hd__o31ai_2        4
     sky130_fd_sc_hd__o32a_2        25
     sky130_fd_sc_hd__o41a_2         1
     sky130_fd_sc_hd__or2_2        178
     sky130_fd_sc_hd__or2_4         19
     sky130_fd_sc_hd__or2b_2        86
     sky130_fd_sc_hd__or3_2        100
     sky130_fd_sc_hd__or3_4          4
     sky130_fd_sc_hd__or3b_2        19
     sky130_fd_sc_hd__or3b_4         3
     sky130_fd_sc_hd__or4_2        147
     sky130_fd_sc_hd__or4_4         58
     sky130_fd_sc_hd__or4b_2         1
     sky130_fd_sc_hd__xnor2_2       22
     sky130_fd_sc_hd__xor2_2        29

   Chip area for module '\top': 213174.451200
```

Open All the Reports with vi/gvim

#Locate the drc and the lvs report

```
/designs/top/runs/RUN_2026.03.24_14.12.30/reports/signoff/drc.rpt
/designs/top/runs/RUN_2026.03.24_14.12.30/reports/signoff/vi
30-top.lvs.rpt
```

Open Both the Reports using vi/gvim editor

```
vi drc.rpt
```

```
top
-----------------------------------------
[INFO]: COUNT: 0
```

```
vi 30-top.lvs.rpt
```

```
LVS reports no net, device, pin, or property mismatches.

Total errors = 0
```

13. APPLICATIONS

The designed RISC-V based SoC subsystem with AXI interconnect is intended for use in lightweight embedded applications such as IoT sensor nodes, smart home devices, industrial monitoring units, and low-power edge-processing systems. The modularity of the RISC-V core combined with AXI-based scalable peripheral integration makes the subsystem suitable for applications requiring programmable control, real-time data acquisition, and efficient communication with external devices.

14. RESULT ANALYSIS

1. . The design successfully meets the target clock period of 20 ns (50 MHz) with no critical setup violations, confirming that it operates reliably at the intended frequency.
2. Hold timing is well-managed, with minor hold issues in early stages resolved by post-CTS resynthesis and tight resizer slack margins, ensuring stable data capture at all sequential elements.

3. Clock Tree Synthesis uses a low skew target and a strong root buffer, effectively balancing the clock network and minimizing clock skew across the design.
4. The synthesis constraints (fanout, transition, capacitance, and output load) are set conservatively, which keeps critical paths robust and avoids over-driving cells.
5. The core area, as derived from the floorplan and density settings, is compact and tightly packed, reflecting efficient use of the available placement region inside the die.
6. The core utilization is set to 35%, which provides sufficient whitespace around cells to ease routing, reduce congestion, and improve timing closure while still maintaining a compact layout.
7. The total chip area is approximately **213,174 µm²**, indicating a relatively small and area-efficient implementation suitable for cost-sensitive or low-power applications.
8. The power grid uses fixed pitch and offset settings for power and ground stripes, giving a predictable distribution network suitable for moderate IR-drop behavior across this chip size.
9. Routing is allowed to use up to Metal4 with sufficient global routing iterations and adjustments, enabling successful completion of both global and detailed routing without major congestion or unrouted nets on this area-constrained chip.
10. Physical verification (DRC, LVS, antenna checks, and basic parasitic extraction) is enabled, ensuring the layout is clean, compliant with technology rules, and ready for sign-off once full power analysis is added for the 213,174 µm² die.

15. PROJECT SUMMARY

- The design was synthesized into a gate-level netlist using Yosys.
- Physical design stages such as floorplanning, placement, clock tree synthesis (CTS), and routing were completed using OpenROAD through the OpenLane automated flow.
- Physical verification including DRC and layout checks was performed using Magic VLSI and KLayout.
- The final GDSII layout file was generated, completing the full RTL-to-GDSII chip implementation flow.

Parameter	Value
Design name	top
PDK / Library	sky130A · sky130_fd_sc_hd
Clock period	20 ns (50 MHz)
Target core util.	35%
Total runtime	0h 47m 46s (routing: 0h 36m 30s)

16. CONCLUSION

In conclusion, the successful physical implementation of this RISC-V SoC demonstrates the effectiveness of the OpenLane open-source flow in transforming a high-level RTL design into a manufacturable GDSII layout. By integrating the PicoRV32 core with AXI-Lite interconnect, ROM, SRAM, and UART, the design achieves a complete and functional system while meeting timing, area, and design rule constraints. The seamless collaboration between tools like Yosys, OpenROAD, OpenSTA and Magic highlights the maturity and reliability of open-source EDA solutions. Overall, this work validates that complex digital systems can be efficiently realized using accessible, cost-effective methodologies, making ASIC design more approachable for research and development.

Overall, the project provides practical understanding of the RTL-to-GDSII chip design process and highlights the usefulness of open-source tools for learning and research in VLSI design.